MAHMOUD AHMED EL-SHARAKY

Software Engineer | EdTech & Learning Platforms

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SUMMARY

Computer Science student (B.Sc., expected 2027) and self-taught software engineer who independently designs, builds, and deploys production systems used by real organizations. Full-stack across relational database design, REST API architecture, and React front-ends, grounded in data structures, algorithms, and software engineering. Shipped an enterprise HR platform serving **600+ employees across 25+ branches** and an EdTech platform with a **13,000+ question** bank. Education technology is the focus: a 13,000+ question bank, many with written explanations, adaptive mock exams, a weakness-analysis engine, and learner analytics — designed, built, and operated solo.

Target role: Software Engineer / Developer in university IT or EdTech — student and staff systems, e-learning and assessment platforms, and academic dashboards.

EXPERIENCE

Full-Stack Developer - SMLE Question Bank - EdTech Platform (Personal)

Feb 2025 - Present | Qassim, SA | smle-question-bank.com

- Built a platform for the Saudi Prometric (SCFHS) exam: a **13,000+ question** bank, adaptive mock tests, and a weakness-analysis engine with real-time analytics.
- Designed the PostgreSQL schema and REST API. Prerendered the public content for search engines: pages indexed by Google rose from **5 to 468** (30 Aug to 16 Sep 2026), with **1,001 visitors** in the last 30 days (Vercel Analytics).
- Built a subscription/payments system on Moyasar (5-tier pricing, Apple Pay, webhook-driven fulfillment, VAT-inclusive invoicing) with a group-purchase flow that mints single-use invite links.
- Shipped a companion mobile app (Expo / React Native) sharing the same backend, alongside a Telegram bot for notifications and admin broadcast.

Full-Stack Software Engineer - Ultimate Care Rehabilitation Co. (Healthcare)

Oct 2025 - Present | Qassim, SA

- Architected and shipped a multi-branch HR platform (~20,000 LOC, 25 relational tables) covering employee lifecycle, payroll, attendance, documents, and transport for **600+ employees across 25+ branches**.
- Implemented JWT authentication with role-based access control (RBAC), TTL-based in-memory caching, and composite database indexing - reaching **sub-second response times** and eliminating N+1 queries.
- Automated document-expiry alerts (30/60/90-day) and Arabic RTL PDF/Excel reporting with Hijri-Gregorian date conversion.

Web Developer (Freelance) - Erth Environmental Services

May 2025 - Jun 2025 | Qassim, SA | erthfc.com

- Built a bilingual (AR/EN) platform with automated consultation workflows and Google Maps integration.

Web Developer (Freelance) - Saleh Al-Hisouni Law Firm

Jul 2025 - Aug 2025 | Qassim, SA | alhisony.com

- Delivered a performant, SEO-optimized website with interactive 3D (Three.js), Schema.org structured data, and custom-domain email for formal correspondence.

PROJECTS

- **Mahfazati (Android, TypeScript, Expo / React Native):** monthly money report built from bank SMS parsed entirely on the phone; 205 automated tests (Vitest); in Google Play closed testing.
- **Central email gateway (Next.js 14, TypeScript, PostgreSQL):** one email quota shared by three production projects, with origin gating, suppression lists and idempotency; 35 automated tests.

EDUCATION

B.Sc. in Computer Science - Qassim University

2022 - 2027 (Expected) | Saudi Arabia

Relevant coursework: Data Structures, Analysis & Design of Algorithms, Operating Systems, Software Engineering, Databases, Logical Design, Coding & Encryption Theory.

TECHNICAL SKILLS

Languages: JavaScript (ES2022+), TypeScript, Python, SQL, HTML, CSS
Backend: Node.js, Express, Next.js, REST API design, JWT / RBAC authentication, Nodemailer
Frontend: React 19, Vite, Framer Motion, Material-UI, Three.js / React Three Fiber
Databases: PostgreSQL (schema design, indexing, query optimization), Supabase, Neon, Redis
Mobile & Payments: Expo, React Native, Moyasar (webhook-driven subscription billing, VAT invoicing)
Tools & Cloud: Git, GitHub, GitHub Actions (CI / scheduled jobs), Vercel (serverless), AWS S3 / Cloudflare R2, Vercel Blob Storage, ESLint
Spoken languages: Arabic (native), English (fluent)

Testing & AI tooling: Vitest, node:test (automated unit tests), AI-assisted development (Claude Code)